BEE: AC Circuits

Page No.
Date

★ Theory: Basic terms & formulae

i) Time period = $\frac{1}{f}$

ii) Angular frequency = $w = 2\pi f$

in qns, phase diff ends up being usually 0, i THINK

iii) Instantaneous value = $v = V_m \sin(wt \pm \phi)$

iv) Equations of alternating current:
→ $i = I_m \sin wt$
→ $e = E_m \sin wt$
→ $v = V_m \sin wt$
} where $wt = \theta$

★ Faraday's law: if a wire cuts through magnetic field lines, a voltage appears in the wire. If the magnetic flux through a coil changes, a voltage appears in the coil.

E_m max. (electromotive force) = $N \cdot B \cdot A \cdot w$ (N: no. of turns, B: flux density, A: area, w: speed)

for a coil with $e = E_m \sin wt$

★ Phase difference & steps for solving reading eqns. problems:
→ ϕ is the phase angle. if ϕ = +ve, left shifted wave
if ϕ = -ve, right shifted wave

→ steps:
1) Write eqn. in the form of $v = V_m \sin(wt \pm \phi)$ or $e = E_m \sin(wt \pm \phi)$

$N \cdot B \cdot A \cdot w$ where $w = \frac{2\pi N_{rpm}}{60}$

2) $f = \frac{w}{2\pi}$ & $T = \frac{1}{f}$

3) for instant. val., substitute t and evaluate in radians.

eg) A 100-turn coil of area $0.02\,m^2$ rotates at 1500 rpm in a uniform field of 0.5T. Find E_m, f, E_{rms} and write instant. eqn.

→ $w = \frac{2\pi N_{rpm}}{60} = \frac{2\pi \times 1500}{60} = 157.08$ rad/s

$f = \frac{157.08}{2\pi} \approx 25$

$\therefore T \approx 0.04$ s

$E_m = N \times B \times A \times w$

$= 100 \times 0.02 \times 0.5 \times 157.08$

$= 157.08$ V

$E_{rms} = \frac{E_m}{\sqrt{2}} = 111.07$ V

$\therefore e = 157.08 \sin(157.08t)$ V

★ Avg. value, RMS value, Form Factor, Crest Factor

→ Since for a symmetrical wave the avg. is zero, we take avg. over a half cycle.

1) $V_{avg} = \frac{1}{\pi} \int_0^{\pi} V_m \sin\theta \, d\theta = \frac{1}{\pi} \times V_m [-\cos\theta]_0^{\pi}$

$\left(\because \text{general } V_{avg} = \frac{1}{T} \int_0^T v\,dt\right) = \boxed{\frac{2}{\pi} V_m} \text{ or } \boxed{0.637\, V_m}$

Page No.
Date

2) Effective V_{rms} : sqrt. of the mean of the squares.

General $V_{rms} = \sqrt{\frac{1}{T}\int_0^T v^2\,dt}$

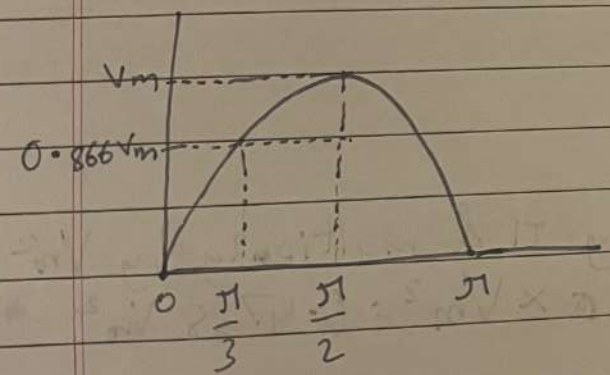
for a sine wave,

$$V_{rms} = \sqrt{\frac{1}{2\pi}\int_0^{2\pi} V_m^2 \sin^2\theta\,d\theta}$$

$$= \sqrt{\frac{V_m^2}{2\pi}\int_0^{2\pi} \frac{1-\cos 2\theta}{2}\,d\theta}$$

$$= \sqrt{\frac{V_m^2}{2\pi} \times \pi} = \sqrt{\frac{V_m^2}{2}} = \boxed{\frac{V_m}{\sqrt{2}}} \text{ or } \boxed{0.707\,V_m}$$

eg) Find RMS of the given waveform.



→ Intervals: $0 - \frac{\pi}{3}$: $v = V_m \sin\theta$

$\frac{\pi}{3} - \frac{\pi}{2}$: $v = 0.866\,V_m$

$\frac{\pi}{2} - \pi$: $v = V_m \sin\theta$

$$\therefore V_{rms}^2 = \frac{1}{\pi}\left(\underbrace{\int_0^{\pi/3} V_m^2 \sin^2\theta \, d\theta}_{(1)} + \underbrace{\int_{\pi/3}^{\pi/2} 0.866^2 V_m^2 \, d\theta}_{(2)} + \underbrace{\int_{\pi/2}^{\pi} V_m^2 \sin^2\theta \, d\theta}_{(3)} \right)$$

(1) $V_m^2 \int_0^{\pi/3} \frac{1-\cos 2\theta}{2} \, d\theta = \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_0^{\pi/3}$

$= \frac{\pi}{6} - \frac{\sqrt{3}}{8} \approx 0.3071$

(2) constant $\therefore (0.866)^2 \times \left(\frac{\pi}{2} - \frac{\pi}{3} \right) \approx 0.3927$

(3) $V_m^2 \int_{\pi/2}^{\pi} \frac{1-\cos 2\theta}{2} \, d\theta = \left[\frac{\theta}{2} - \frac{\sin 2\theta}{4} \right]_{\pi/2}^{\pi}$

$= \frac{\pi}{2} - \frac{\pi}{4} \approx 0.7854$

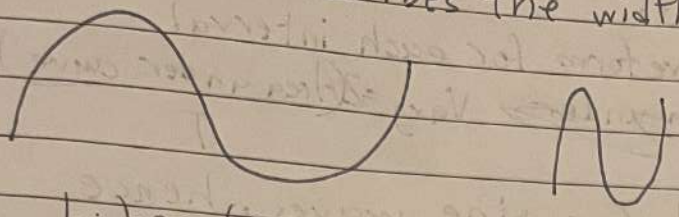
Add (1), (2), (3) and divide by π & multiply by V_m^2

$\frac{0.3071 + 0.3927 + 0.7854}{\pi} \times V_m^2 = 0.4728 \, V_m^2$

Take sqrt.

$\boxed{0.688 \, V_m}$

3) Form factor: Describes the width of a wave.



higher form factor

lower form factor

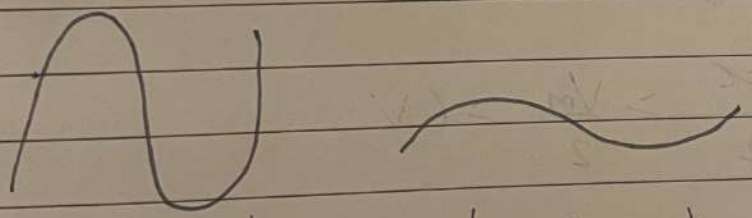
Form factor = $\frac{V_{rms}}{V_{avg}}$

For a sine wave = $\frac{0.707 V_m}{0.637 V_m} = 1.11$

Other waves:

1) Half-wave rectified = 1.57
2) Square wave = 1
3) Triangular wave = 1.155

4) Crest factor: Describes the peaks of the wave, whether they're sharp or not.



higher crest factor

lower crest factor

for a sine wave, $V_{peak} = V_{max}$

Crest factor = $\frac{V_{peak}}{V_{rms}} = 1.414$

Other waves:

1) Half-wave rectified = 2
2) Square wave = 1
3) Triangular wave = 1.732

→ steps for problems:

1) Write eqn. of waveform for each interval
2) ~~Average~~ V_{avg} (area under) $V_{avg} = \frac{\text{(Area under curve)}}{T}$

Use half a period for sine waves, hence 2×Area. Otherwise, use full period.

3) For RMS, square the funct, integrate over the period and divide by T, (T/2 if symm. function) then take sqrt.

eg) Non-sine eg.
A ~~triangular~~ sawtooth wave rises linearly from 0 to 12V over one period T and then drops instantly to 0. Find V_{avg}, V_{rms}, form factor, crest factor.

→ $V_{avg}: \frac{1}{T}\int_0^T V\,dt$

$\therefore V_{avg} = \frac{1}{T}\int_0^T \left(\frac{V_m}{T}t\right)dt$

$= \frac{V_m}{T^2} \times \frac{T^2}{2} = \frac{V_m}{2} = 6V$

→ $V_{rms}^2: \frac{1}{T}\int_0^T V^2\,dt$

$\therefore V_{rms}^2 = \frac{1}{T}\int_0^T \frac{V_m^2}{T^2}t^2\,dt$

$= \frac{V_m^2}{T}\int \frac{t^2}{T^2}\,dt$

$= \left(\frac{V_m^2}{T^3}\right)\left(\frac{T^3}{3}\right) = \frac{144}{3} = 48 \quad \therefore V$

Page No.
Date

Form factor = $\frac{6.93}{6} = 1.155$

Crest factor = $\frac{12}{6.93} = 1.732$

✡ Phasors:

Suppose we have 2 sources in series
$V_1 = 10\sin\omega t$ and $V_2 = 10\sin(\omega t + 90°)$
to form phasors, take the real part as either:

i) Rectangular form: $V = a + jb$ (real, imaginary, $j = \sqrt{-1}$)

ii) Polar form: $V = |V|\angle\phi$

Using rectangular form,
$A = v\cos\phi$ and $B = v\sin\phi$
for V_1

$v\cos\phi = 10 \times \cos 0°$
$= 10$

$v\sin\phi = 10 \times \sin 0°$
$= 0$

for V_2

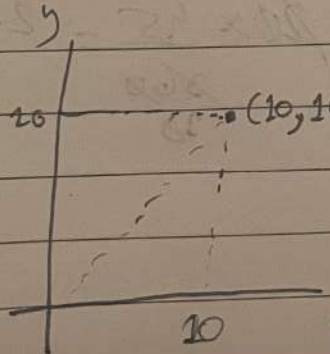
$v\cos\phi = 10 \times \cos 90°$
$= 0$

$v\sin\phi = 10 \times \sin 90°$
$= 10$

$\therefore V = 10 + j10 \longrightarrow$

$\therefore$ length $= \sqrt{100 + 100}$
$= 14.14$

$\therefore$ angle $= \tan^{-1}\left(\frac{10}{10}\right) = 45°$



$\therefore$ Result $= 14.14\sin(\omega t + 45°)$

⚠ Using V_{max} here for sake of convenience the standard is to use V_{RMS}.

→ Converting from polar to rectangular:
$a = V\cos\phi$, $b = V\sin\phi$

→ Converting from rectangular to polar.
$|V| = \sqrt{a^2 + b^2}$, $\phi = \tan^{-1}\left(\frac{b}{a}\right)$

→ Current lags if its angle is smaller. Current leads if its angle is larger.

eg) $v = 141.4\sin(314t + 30°)$
$i = 7.07\sin(314t - 15°)$
find f, rms vals, phase difference, which one leads, and time gap between their peaks.

→ $V_{rms} = \frac{141.4}{\sqrt{2}} = 100\ V$

$I_{rms} = \frac{7.07}{\sqrt{2}} = 5\ A$

$\phi = 30 - (-15) = 45°$
∴ Voltage leads by 45°

$f = \frac{\omega}{2\pi}$, $\frac{314}{2\pi} = 50\ Hz$ ∴ $T = \frac{1}{50} = 20\ ms$

Time gap $= 20 \times \frac{45}{360} = 2.5\ ms$